Cross-Patient Neural Speech Decoding

Ben Tang | Cogan Lab, Duke University | April 2026

1 What Is Shared Across Patients?

Conserved cortical organization

Speech motor cortex has a **conserved somatotopic layout** across all humans: face, jaw, tongue, and larynx representations are arranged in a consistent order along the central sulcus (Breshears 2015, n=33; Bouchard 2013; Roux 2020). The *ordinal* arrangement is invariant; the *absolute positions* vary by 5–15 mm between individuals.

We define 16 **cortical nodes** at atlas-derived positions (Brainnetome, N=40 HCP subjects) corresponding to known speech-relevant regions. These are the subject-invariant functional units:

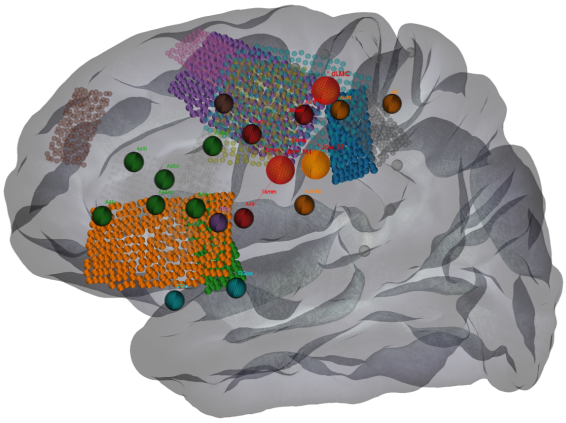

Figure 1: 16 cortical nodes (large spheres) and patient electrode arrays (small dots). Each patient covers a different subset of nodes.

Key nodes along the motor strip (dorsal → ventral): **A4hf** (face M1, MNI −49/−6/40) → **A6cvl** (ventral PMC, −49/6/31) → **A4tl** (tongue M1, −52/2/9). Plus 3 sensory (face/tongue S1, proprioceptive), 6 Broca's (articulatory planning), 2 auditory (self-monitoring), 1 insula, 1 prefrontal. Each node reachable by 3–9 of our 10 patients.

Subject-invariant phoneme dynamics

Each phoneme activates a **specific, predictable combination of cortical nodes** determined by articulatory biomechanics:

- **/b/, /p/** (bilabial stops): lip closure → **face M1** fires
- **/g/, /k/** (velar stops): tongue dorsum raises → **tongue M1** fires
- **/v/** (labiodental fricative): lower lip to teeth → **face M1** fires
- **/a/** (open vowel): jaw opening + tongue low → **tongue M1** + jaw area
- **/i/** (high front vowel): tongue raised front → **tongue M1**

- **/u/** (high back vowel): tongue raised back + lip rounding → **tongue M1 + face M1**

These node activations follow a conserved temporal cascade: **Broca's** (-200 to -50 ms, planning) → **Motor cortex** (-50 to $+100$ ms, execution) → **Somatosensory** ($+50$ to $+200$ ms, feedback) → **Auditory** ($+100$ to $+300$ ms, self-monitoring).

The ground truth: what is conserved across patients is (1) the existence and gross positions of cortical nodes representing articulatory muscles, and (2) the spatiotemporal activation sequences across these nodes during phoneme production. A 3-phoneme token like /abe/ produces a characteristic trajectory through the node network — Broca's plans the sequence, then face M1 fires for /b/, tongue M1 for /a/ and /e/, with sensory and auditory feedback following each.

2 What Do We Observe?

We have **noisy, spatially-sampled observations** of these node activations. Each patient has a μECoG electrode array (128 or 256 channels, 1.3–1.7 mm pitch) pressed against the cortical surface during intra-operative speech. We measure high-gamma activity (HGA, 70–150 Hz envelope at 200 Hz) — a proxy for local cortical population activity.

The physics of this measurement is well-characterized: the signal at each electrode is a **linear superposition** of cortical sources, contributions **decay with distance** ($\sim 1/d^2$), and sources beyond $\sim$25 mm are below the noise floor. Crucially, this spatial mapping depends **only on geometry** — it does not change based on what the brain is currently doing.

Every patient's array sits in a **different position** (15–25 mm offsets, variable rotation). Each array is a partial view of the cortical node network from a different vantage point.

3 What Must We Learn?

Per patient (geometry — *where* things are)

1. **Electrode localization error** (6 params): Surgical reconstruction has $\sim$4–8 mm systematic rigid-body error. Learn translation Δ + rotation ω, clamped ($\|\Delta\| \leq 15$ mm, $\|\omega\| \leq 0.15$ rad).

2. **Individual cortical node positions** (48 params): Individual anatomy varies by 5–15 mm from atlas. Each of 16 nodes gets a learned offset $\delta_l \in \mathbb{R}^3$, clamped $\leq$15 mm. Initialized at zero (trust the atlas).

3. **Signal gain and offset** (128 params): Electrode impedance varies. Per-channel scale (positive via softplus) and offset. Initialized from channel statistics.

Total: 182 parameters per patient. Comparable cross-patient iEEG systems (seegnificant, MIBRAIN, BarISTA) use per-subject heads or region prototypes with similar or larger per-patient footprints, or no per-patient adaptation at all.

Shared (dynamics — *what* things mean)

Spatiotemporal activation patterns on cortical nodes → which phoneme sequence. Given correctly-positioned nodes with correctly-calibrated signals pooled into them, the 16-node network dynamics encode the phoneme. Same mapping for every patient — determined by conserved cortical organization, not individual anatomy.

4 What Architecture Does This Ask For?

Each component follows directly from the physics:

```
Raw electrode HGA (N_i electrodes x T time steps)
  -> Temporal binning: Conv1d(1->d, k=10, s=10)        [200Hz -> 20Hz]
  -> Per-patient calibration:                           [182 params/pt]
       softplus(s) * x + b   (gain/offset, positive)
       correct electrode positions (6 params: rigid)
       correct node positions (48 params: individual somatotopy)
  -> Spatial pooling into 16 cortical nodes             [physics-derived]
       Wendland kernel, distance-weighted, compact support
  -> [Node self-attention + Temporal self-attention] x 2  [learns dynamics]
  -> AR decoder: predict phoneme 1, 2, 3 + beam search   [52 valid tokens]
```

Spatial pooling

Each node computes a weighted average of nearby electrode signals, where the weights depend only on distance — electrodes closer to the node contribute more, and electrodes beyond a learned radius R contribute nothing:

$$z_{\text{node}} = \sum_j \underbrace{\frac{w(\text{dist}_j,\, R)}{\sum_{j'} w(\text{dist}_{j'},\, R)}}_{\text{distance-only weights}} \cdot \underbrace{(x_j\, W)}_{\text{learned features}} \tag{1}$$

Multiple pooling heads operate at different radii (tight ∼10 mm for focal activity, broad ∼25 mm for regional), each extracting different learned features before averaging.

Parameter budget

Component	Params	Note
Conv1d + feature projections	∼5K	Shared: temporal features
Spatial pooling (Wendland)	H scalars	Shared: learned R_h per head
Spatiotemporal blocks (×2)	∼132K	Shared: node dynamics
AR decoder	∼4K	Shared: phoneme sequence
Gain/offset (softplus$(s) \odot x + b$)	128/pt	Per-patient: impedance
Electrode correction (Δ, ω)	6/pt	Per-patient: registration error
Node offsets (δ_l)	48/pt	Per-patient: individual somatotopy
Total shared	**∼141K**	
Total per-patient	**182**	

Immediate next steps

1. **Scale data** (highest priority) — to learn the shared spatiotemporal dynamics we need a foundation of data. Could request chronic ECoG speech data from Flinker/Chen (NYU, 48 patients) and/or Chang (UCSF, ∼15–25 patients)
2. **Nonlinear MNI normalization** — current electrode transform is linear-only (∼5–15 mm residual error); if T1 MRIs exist, ANTs SyN could halve this
3. **Implementation** — architecture is designed; target ∼2 weeks to first training run on DCC